

$$415 \div 23 = 18 \text{ R } 5$$
$$p = 18$$
$$r = 5$$

V) 5.1)

a) Signed Zahlenbereiche:

b) $\left[-\frac{b^n}{2}, \frac{b^n}{2} - 1 \right]$

d) 5.2) n: Anzahl der Stellen

a) $10_{10} \cdot 10_{10}^4 = 4355_{10} = 05645_{10}, n = 4$

$10_{10} \cdot 10_{10}^4 = 7286_{10} = 02714_{10}, n = 4$

$10_{10} \cdot 10_{10}^3 = 2714_{10} = 0104_{10}, n = 3$

$$\begin{array}{r} 1000 \\ \text{elc} \\ \hline 104 \end{array}$$

$10_{10} \cdot 10_{10}^3 = 800_{10} = 0800_{10}, n = 3$

$10_{10} \cdot 10_{10}^5 = 10010_{10} = 01110_{10}, n = 5$

$$\begin{array}{r} 1000000 \\ - 10010 \\ \hline 001110 \end{array}$$

$10_{10} \cdot 10_{10}^5 = 10001_{10} = 01110_{10}, n = 5$

$$\begin{array}{r} 1000000 \\ - 10001 \\ \hline 001111 \end{array}$$

b) $4355_{10} = +4355$

$7286_{10} = 7286_{10} - 10000_{10} = -2714_{10}$

$2714_{10} = 2714_{10} - 10000_{10} = -104_{10} = -260$

$800_{10} = -800_{10} = 6$

$010010_2 = +10010_2 = +18_{10}, n = 6$

$10001_2 = -10001_2 = -17_{10}, n = 5$

c) $3 \cdot 16^3 + 10 \cdot 16^2 + 3$

$04355_{10} \ll 2 = 35500_{10}$

5.5

a) $\left(43550, 72860, 7770, 27140, 8000, 0100100 \right)$

b) 10

$04355_{10} \gg 2 = 00043$

$7286_{10} \gg 2 = 0072$

$777_{10} \gg 2 = 007$

$2714_{10} \gg 2 = 00e$

$800_{10} \gg 2 = 00e$

010010_2

10001_2

c) arithmetisch

$04355_{10} \gg 2 = 00043$

$7286_{10} \gg 2 = 0072$

$777_{10} \gg 2 = 007$

$2714_{10} \gg 2 = 00e$

$800_{10} \gg 2 = 008$

$010010_2 \gg 2 = 00100$

$10001_2 \gg 2 = 11100$

b) $415_{10} = (100 \cdot 10^3 + 1 \cdot 10^2 + 100 \cdot 10^1 + 10^0)_2$

$= 100 \cdot 1100100 + 1010 + 1001 + 1001$

$= 1100100100 + 1010 + 1001 + 1001$

$= 1100100100101010101$

$23_{10} = 20 \cdot 3$

$23_{10} = (10 \cdot 1010 + 11)_2$

$23_{10} = 10111$

$$\begin{array}{r} 1101000111 + 10111 = 100110 \\ - 10110 \\ \hline 11001 \end{array}$$

$100110_2 = 18_{10}$

$1011_2 = 5_{10}$

$$\begin{array}{r} 4254567267 \div 641 = 6 \end{array}$$

5.3

a) $0001, 53f5, 7ff, 8000, 5003, fff$

b) 5003

$$\begin{array}{r} 5003 \\ + 53f5 \\ \hline 88f8 \end{array}$$

5003 Carry: 3

$7ff$ Overflow: 3

$$\begin{array}{r} 5003 \\ + 7ff \\ \hline 11602 \end{array} \Rightarrow 1602$$

$7ff$ Carry: 1

$$\begin{array}{r} 7ff \\ + 8000 \\ \hline fff \end{array}$$

5.3

a) $0001, 53f5, 7ff, 8000, 5003, fff$

b) 5003

$$\begin{array}{r} 5003 \\ + 53f5 \\ \hline 88f8 \end{array}$$

5003 Carry: 3

$7ff$ Overflow: 3

$$\begin{array}{r} 5003 \\ + 7ff \\ \hline 11602 \end{array} \Rightarrow 1602$$

$7ff$ Carry: 1

$$\begin{array}{r} 7ff \\ + 8000 \\ \hline fff \end{array}$$

VI)

a) $ca5f_{16}$

$$\begin{array}{r} 1100000000000000 \\ 1010000000000000 \\ 1001000000000000 \\ \hline 1111 \end{array}$$

$1100, 1010, 1001, 1111$

$ca5f_{16} = 0110 0000 0000 0000$

$0110 0000 0000 0000$

$1101 0000 0000 0000$

$1100 0000 0000 0000$

1111

6.3

a)

$0 \times 0000 = 0000 0000 0000 0000$

$0 \times ca5f = 1100 1010 1001 1111$

$0 \times abab = 1010 1011 1010 1011$

$0 \times edda = 1110 1101 1101 1010$

$0 \times ffff = 1111 1111 1111 1111$

$\rightarrow$ code

6.4

$0 \times ca5f = 1100 1010 1001 1111$

$y: 0111 0100 0001 0001$

$y \Rightarrow 0x7411$

$0 \times b2ef = 1011 1110 1110 1111$

6.2

$y: (x \gg k) \Rightarrow \underbrace{0 \dots 0}_k x_u$

$x \ll (n-k) = x_{n-k} \dots \underbrace{0 \dots 0}_{n-k}$

$$\underline{y} = \underbrace{0 \dots 0}_k x_u \mid x_{n-k} \dots \underbrace{0 \dots 0}_{n-k}$$

$$\underline{y} = \left\{ \underbrace{(0 \mid x_{n-k}) \dots (x_{n-k} \mid 0)}_k \right\}$$

a)

$1100 1011 1001 1111$

$0110 1101 1100 1111$

$\Rightarrow 0100 1001 1000 1111$

c)

$1100 1011 1001 1111$

$0110 1101 1100 1111$

$\Rightarrow 1110 1111 1101 1111$

d)

$1100 1011 1001 1111$

$0110 1101 1100 1111$

$\Rightarrow 1010 0110 0101 0000$

e)

$1100 1011 1001 1111$

$0110 1101 1100 1111$

$\Rightarrow 1011 0110 0111 0000$

f)

$1100 1011 1001 1111$

$0110 1101 1100 1111$

$\Rightarrow 0001 0000 0010 0000$

b) $0 \times a6a6 = 1010 1010 1010 1010$

$0 \times 5555 = 0101 0101 0101 0101$

c) $0 \times ffff = 1111 1111 1111 1111$

$0 \times 7411 \Rightarrow ffff$

$0 \times 0000 = 0000 0000 0000 0000$

d) 1467

$0 \times dea2 = 1101 1110 1010 1101$

$y \Rightarrow 0001 0100 0110 0111$

$\Rightarrow 1467$

$0 \times a6a6 = 1100 1010 1100 1010$

6.5a)

1. $\underline{y} = 1111$

2. $\underline{y} \ll 26$

3. $\underline{z} \leftarrow \underline{z} \& (\underline{y} \sim \underline{z})$

4. $\underline{z} \leftarrow \underline{z} \mid \underline{y}$

6.5b)

$\underline{z} \leftarrow (\underline{z} \ll 3) \gg 26$

$\underline{y} \leftarrow (\underline{y} \ll 28) \gg 28$

$\underline{y} \leftarrow \underline{y} \mid \underline{z}$

7.1a)

$2^{-1} = \frac{1}{2} = 5 \text{ E} - 1_{10} \rightarrow 1 \text{ E} - 1$

$2^{-2} = \frac{1}{4} = 25 \text{ E} - 2_{10} \rightarrow 1 \text{ E} - 10$

$2^{-3} = \frac{1}{8} = 125 \text{ E} - 3_{10} \rightarrow 1 \text{ E} - 100$

$2^{-4} = \frac{1}{16} = 0625 \text{ E} - 4_{10} \rightarrow 1 \text{ E} - 1000$

$2^{-5} = \frac{1}{32} = 03125 \text{ E} - 5_{10} \rightarrow 1 \text{ E} - 10000$

$2^{-6} = \frac{1}{64} = 015625 \text{ E} - 6_{10} \rightarrow 1 \text{ E} - 100000$

6.5)

a)

i) $x_{28} - x_{26} \Rightarrow [0, 7]$

ii) $\Rightarrow [-4, 3]$

$x_{23} - x_{21} : \dots$

$x_{10} - x_8 : i) [0, 15]$

ii) $[-8, 7]$

b) $m = \{1, 1, 1, \dots, 0, 0, 0, \dots\}$

Schritt die man auf 0 setzen muss

32

Bitweise: $\underline{z} \& \underline{m} \rightarrow \underline{z}$

c) $0 \times f2edcace = 1111 0010 1110 1101 1100 1010 1100 1110$

$x_7 - x_6$ $x_5 - x_4$ $x_3 - x_2$

i) $100_2 = 4_{10}$ i) $111_2 = 7_{10}$ i) $0101_2 = 5_{10}$

ii) $= -1_{10}$ ii) $= -4_{10}$ ii) $= 5_{10}$

7.2

i)

A $18 \rightarrow 12 : 1$

B $30 \rightarrow 36 : 3$

C $20 \rightarrow 24 : 2$

D $34 \rightarrow 36 : 3$

E $18 \rightarrow 12 : 1$

10 Autos $\Rightarrow 3a$

ii)

A $18 \rightarrow 24 : 2$

B $30 \rightarrow 36 : 3$

C $20 \rightarrow 24 : 2$

D $34 \rightarrow 36 : 3$

E $18 \rightarrow 24 : 2$

12 Autos $\Rightarrow$ kein

b) $4375_{10} = 4375 \cdot 1 \Rightarrow 100010001011$

$2187 \cdot 1 = 43.75 = 10101111$

$1053 \cdot 1 = 4.375 = 100.011$

$546 \cdot 0 = 0.4375 = 0.0111$

$273 \cdot 1 = 0.04375 = 0.0001$

$136 \cdot 0$

$68 \cdot 0$

$34 \cdot 0$

$17 \cdot 1$

$8 \cdot 0$

$4 \cdot 0$

$2 \cdot 0$

$1 \cdot 1$

7.3)

i) $5.314_{10} \cdot 10^2 = 531.4_{10}$

$[531.4 + 0.8500] \cdot 10^2$

$= 532.25$

ii) $1.740_{10} \cdot 10^2 = 174.0_{10}$

$(1.740 + 6.375) \cdot 10^2$

$10.00125 \cdot 10^3$

$= 10001.25$

iii) $2.1_{10} \cdot 10^2 / 5.1_{10} \cdot 10^3$

$210 / 0.00051$

411764.7058

iv) $1.011_{10} \cdot 2^5 + 1.100_{10} \cdot 2^4$

$101100 + 110000$

$= 1000100_2$

8.1a) AND, NOR, NAND, XOR

b) OR, NOR, NAND

c) i)

a	b	c	a+b	(a+b)c	ac	bc	ac+bc
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	1	0	1	1
1	0	0	1	0	0	0	0
1	0	1	1	1	1	0	1
1	1	0	1	0	0	0	0
1	1	1	1	1	1	1	1

iii) → Gült

a	b	c	a ⊕ b	(a ⊕ b)c	ac	bc	ac ⊕ bc
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	1	0	1	1
1	0	0	1	0	0	0	0
1	0	1	1	1	1	0	1
1	1	0	0	0	0	0	0
1	1	1	0	0	1	1	0

→ gilt nicht

iv)

a	b	c	a NOR b	NAND c	a NAND b	b NAND c	NOR
0	0	0	1	1	1	1	0
0	0	1	1	0	1	1	0
0	1	0	1	1	1	0	0
0	1	1	0	1	1	0	0
1	0	0	0	1	1	1	0
1	0	1	0	0	1	1	0
1	1	0	0	1	1	1	0
1	1	1	0	1	0	0	1

→ es gilt nicht

7.4)

i) [-5000, 4999] iii) [-80, 75]

ii) [-435, 500] iv) [-011111111111, 111000000000]

7.5)

a = 0x9b c = 0x63

b = 0xD5 d = 0x25

b) 2c)

e = 0x15 = 76₁₀

f = 0x7f = 127₁₀

g = 0x0a = 10₁₀

h = 0xd2 = 210₁₀

9.1)

a)

Beispiel 1:

$$\begin{matrix} a & = & 1 \\ b & = & 0 \\ x & = & 1 \end{matrix}$$

$$a + x = b + x$$

$$\Rightarrow 1 + 1 = 0 + 1 \Rightarrow 1 = 1$$

jedesmal: 1 ≠ 0, a + b

Beispiel 2: a := z + y

$$\begin{matrix} b := z \\ z := 1 \\ x := 1 \\ y := 0 \end{matrix}$$

⇒ a · x = b · x

$$(z + y) · x = z · x$$

$$1 · 1 = 1 · 1$$

$$1 = 1$$

jedesmal: z + y ≠ z, a ≠ b

ii)

a	b	c	ab	ab+c	a+c	b+c	(a+c)(b+c)
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	0	0	1	1	0
0	1	1	0	1	1	1	1
1	0	0	0	0	1	0	0
1	0	1	0	1	1	1	1
1	1	0	1	1	1	1	1
1	1	1	1	1	1	1	1

→ gilt nicht

a	b	c	bc	a → (bc)	a → b	a + c	(a → b)(a + c)
0	0	0	0	1	1	1	1
0	0	1	0	1	1	1	1
0	1	0	0	1	1	1	1
0	1	1	1	0	1	1	1
1	0	0	0	1	0	0	0
1	0	1	0	1	0	1	0
1	1	0	0	1	1	0	0
1	1	1	1	1	1	1	1

8.5a)

x	x ₈	y	y ₈	z	z ₈
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	0	0
0	0	1	1	0	1
0	1	0	0	1	0
0	1	0	1	1	1
0	1	1	0	1	0
0	1	1	1	1	1
1	0	0	0	1	0
1	0	0	1	1	1
1	0	1	0	1	0
1	0	1	1	1	1
1	1	0	0	0	0
1	1	0	1	0	1
1	1	1	0	0	0
1	1	1	1	0	1

7.6

a)

i) 14,375₁₀ = 1110,011₂

$$1,110011_2 \cdot 10^3$$

$$= 0,10000011 \cdot 110011$$

$$\Rightarrow 0,1000001101100110$$

32 bit

ii) -173,875₁₀ = -1010,1101,111

64 bit ⇒ 1 100 0000 0110 0101 1011 1100 0000 ...

iii)

$$0,03125_{10} = 0,00001_2$$

32 bit ⇒ 0 0111 1010 0000 ...

iv)

$$-0,046875_{10} = -0,000011$$

signed, 32 bit ⇒ 1 0111 1010 1000 0000 ...

signed, 64 bit ⇒ 1 0111 1111 1010 1000 0000 ...

9.1b)

$$((a + x = b + x) \wedge (a + x' = b + x')) \Rightarrow a = b$$

wenn (a + x = b + x) ∧ (a + x' = b + x')

folgt: (a + x' = a + x) ∧ (b + x = b + x') (Kommutativgesetz)

daraus folgt, dass x trivial für den Wert von A ∧ B ist. Daraus folgt:

$$(a = a) \wedge (b = b)$$

daraus folgt, dass a = 1 & b = 1

⇒ a = b

Das gleiche gilt für die 2n. Gleichung, nur dass hier x = 1 sein muss.

9.2)

a)

i) xxy' + xxxy' + yz = xy' + yz

ii) (x + y')(x' + xy'z)

$$= (x + y')(x' + y)(x' + z)$$

iii) (x + y'z)(y + z)

$$(x + y')(x + z)(y + z)$$

iv) (x'y')(x' + y) = x'y'

v) xy' + y'z = y'(x + z)

vi) yx'(y + z)

9.2b)

i) ((x' + y)(x' + y + z))'

ii) (x + y')(x' + xy'z)

$$= (x + y')(x' + y)(x' + z)$$

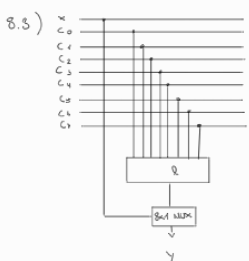
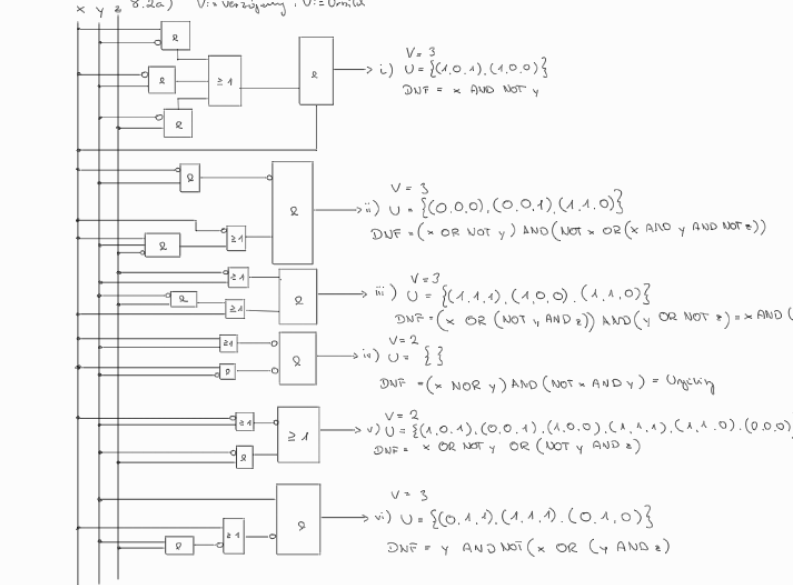
iii) (x + y'z)(y + z)

$$(x + y')(x + z)(y + z)$$

iv) (x'y')(x' + y) = x'y'

v) xy' + y'z = y'(x + z)

vi) yx'(y + z)

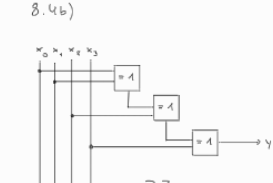
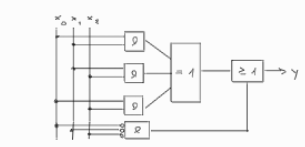


U = {(1,1,0,1)}

x	c ₂	c ₁	c ₀	y
1	1	0	1	x ⇒ 1
0	1	0	1	x ⇒ 0
0	1	1	1	...
...	...	...	...	...

8.4a)

$$U = \{(1,0,1), (1,1,0), (0,1,1)\}$$

$$DNF = (x \text{ AND } y) \text{ XOR } (y \text{ AND } z) \text{ XOR } (x \text{ AND } z)$$


7.7

$$f = 0,100101110000 \dots$$

$$i = 0,100101110000 \dots$$

unterschied zu bin, 1000 zu 1001

9.3)

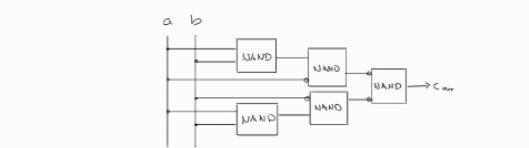
a) zu zeigen: a ⊕ b = (a(b')'(b(ab'))')'

Definition: a ⊕ b = a'b + ab'

$$((a(b')'(b(ab'))')')'$$

$$= a(ab)' + b(ab)'$$

$$= a(a' + b') + b(a' + b)$$

$$= ab' + ba' = a'b + ab'$$


b)

a ⇒ b | Definition von "⇒"

a' + b | ' = ¬, alles wird invertiert

$$7(a'b') \Leftrightarrow (ab)'$$

1. a ⊕ b = (a+b)(a' + b')

2. (a + b) ⊕ b'

$$\Rightarrow ((a+b) + b')((a+b)' + b)$$

$$= a'b' + b = a' + b = (ab)'$$

$$\Rightarrow ((c + b) \oplus b') = (cb)'$$

1. ((a+b) ⊕ b)'

2. (a ⊕ b)' = a'b + ab

$$\Rightarrow (a + b)b' + (a + b)b = ab' + ab + ab + bb$$

$$= a'b' + b = a' + b \quad (ab)'$$

5.5

b)

i) $x \cdot xy' + xy' \cdot y + y'z = xy' + y'z \Rightarrow x(y \oplus 1) + (y \oplus 1)z = xy \oplus x + zy \oplus z$

ii) $(x + y')(x' + xy'z)$
 $= xx' + xxy'z + y'y' + y'xy'z$
 $= xy'z + y'y' \Rightarrow xy'z + (y \oplus 1)(x \oplus 1)$

iii) $xy + y'zy + xz' + yz'$
 $= xy + xz' \Rightarrow xy + xz \oplus x$

iv) $x'y' \cdot (x' + y)$
 $= x'x'y' + x'y'y$
 $= x'y'y = (x \oplus 1)(y \oplus 1)$

v) $xy' + y'z = xy \oplus x + zy \oplus z$

vi) $yx'(y' + z) = yx'y' + yx'z = yx'z = yx'z \oplus yz$

10.1

a)

	00	01	11	10
0	1		1	1
1		1		

3D

	00	01	11	10
00				
01				
11				
10				

4D

$x_4=0$	00	01	11	10
00				
01				
11				
10				

$x_4=1$	00	01	11	10
00				
01				
11				
10				

5D

$x_1x_2=00$	00	01	11	10
00				
01				
11				
10				

$x_1x_2=01$	00	01	11	10
00				
01				
11				
10				

$x_1x_2=11$	00	01	11	10
00				
01				
11				
10				

$x_1x_2=10$	00	01	11	10
00				
01				
11				
10				

6D

b)

3D: Kanten: $\binom{3}{1} \cdot 2^{3-1} = 12$
 Flächen: $\binom{3}{2} \cdot 2^{3-2} = 6$

4D: Kanten: $\binom{4}{1} \cdot 2^{4-1} = 32$
 Flächen: $\binom{4}{2} \cdot 2^{4-2} = 24$

5D: Kanten: $\binom{5}{1} \cdot 2^{5-1} = 80$
 Flächen: $\binom{5}{2} \cdot 2^{5-2} = 80$

6D: Kanten: $\binom{6}{1} \cdot 2^{6-1} = 192$
 Flächen: $\binom{6}{2} \cdot 2^{6-2} = 240$

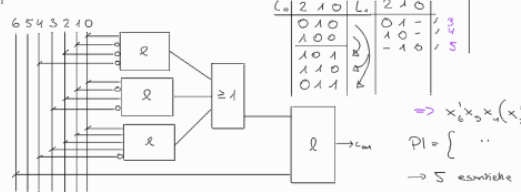
10.5)

	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
0000																
0001																
0010																
0011																
0100																
0101																
0110																
0111																
1000																
1001																
1010																
1011																
1100																
1101																
1110																
1111																

	6	5	4	3	2	1	0	C_{err}
1	0	0	0	0	0	1	1	1
1	0	0	0	1	0	1	1	1
1	0	0	1	0	0	1	1	1
1	0	0	1	1	1	1	1	1
1	0	1	0	1	0	1	1	1
1	1	0	0	0	0	1	1	1
1	1	0	1	0	0	1	1	1
1	1	0	1	1	1	1	1	1
1	1	1	0	1	0	1	1	1

McCluskey:

L_0	4	3	2	1	0	L_1	4	3	2	1	0
	0	0	0	0	1		0	-	0	0	1
	0	0	1	0	1		-	0	1	0	1
	0	1	1	1	1		0	1	1	1	1
	1	0	1	0	1						



$\Rightarrow x_6(x_5x_4x_3 + x_5x_4x_2 + x_5x_3x_2 + x_4x_3x_2 + x_4x_2x_1 + x_3x_2x_1)$

$\Rightarrow 5$ essentielle Prinzipalkombinationen

Essentielle PI: $\{x_6x_5x_4x_3x_2, x_6x_5x_4x_3x_1, x_6x_5x_4x_2x_1, x_6x_5x_3x_2x_1\}$

$f'(A) = \{ \text{alle Variable in bin } 7 \}$

1. $(b' \Rightarrow a')$

2. $(a \Rightarrow b) = (a' + b)$ (Definition)

$\Rightarrow (b' \Rightarrow a') = (b + a') = (a' + b)$

22.: $((a \Rightarrow b)(b \Rightarrow c))' = (a \oplus b)$

1. $(a \oplus b) = ab' + a'b$ (Definition)

2. $(a \Rightarrow b) = a' + b$ (Definition)

$\Rightarrow ((a' + b)(a' + b'))' = ab' + a'b$

$(a' + b)' + (a' + b')' = \dots$

$ab' + a'b = a'b + a'b'$

d)

22.: $((a \Rightarrow b)(b \Rightarrow c)) \Rightarrow (a \Rightarrow c) = 1$

1. $(a \Rightarrow b) = (a' + b)$ | Definition

$\Rightarrow (((a' + b)(b' + c))' + (a' + c)) = 1$

$((a' + b)' + (b' + c)' + (a' + c)) = 1$

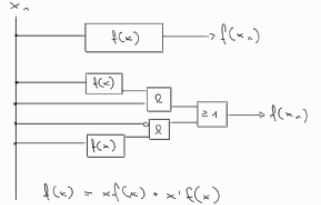
$(ab' + bc' + a' + c) = 1$

$(b'a' + a') + (bc' + c) = 1$

$b' + b = 1$

g.h)

a)



für x folgt f(x)

für nicht x folgt f(x)

somit ist x in "x f(x) + x' f(x)" inkl.

$\Rightarrow x f(x) + x' f(x) = f(x)$

				1	2	3	4	
a	b	c	d	$a \oplus b$	$(a' + c)$	$c \oplus b'$	$b' + 3$	$2 \oplus 4$
0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0
0	0	1	0	0	0	0	0	0
0	0	1	1	0	0	1	0	0
0	1	0	0	1	1	0	0	1
0	1	0	1	1	1	0	0	0
0	1	1	0	1	0	0	0	0
0	1	1	1	1	0	1	1	1
1	0	0	0	1	1	0	0	1
1	0	0	1	1	1	0	0	0
1	0	1	0	1	0	0	0	0
1	0	1	1	1	0	1	0	0
1	1	0	0	0	0	0	0	0
1	1	0	1	0	0	0	0	0
1	1	1	0	0	0	0	0	0
1	1	1	1	0	0	1	1	1

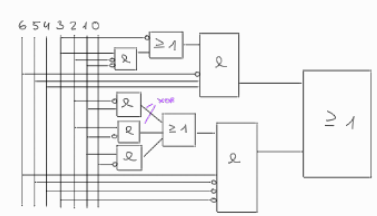
1

2

	6	5	4	3	2	1	0	C_{err}
0	1	1	0	0	0	0	1	1
0	1	1	0	0	0	1	1	1
0	1	1	0	0	1	0	1	1
0	1	1	0	0	1	1	1	1
0	1	1	0	1	0	1	1	1
0	1	1	0	1	1	0	1	1
0	1	1	0	1	1	1	1	1
0	1	1	1	0	0	0	1	1
0	1	1	1	0	0	1	1	1
0	1	1	1	0	1	0	1	1
0	1	1	1	0	1	1	1	1
0	1	1	1	1	0	0	1	1
0	1	1	1	1	0	1	1	1
0	1	1	1	1	1	0	1	1
0	1	1	1	1	1	1	1	1
1	0	0	0	0	0	1	1	1
1	0	0	0	0	1	0	1	1
1	0	0	0	1	0	0	1	1
1	0	0	0	1	0	1	1	1
1	0	0	1	0	0	0	1	1
1	0	0	1	0	0	1	1	1
1	0	0	1	1	0	0	1	1
1	0	0	1	1	0	1	1	1
1	0	0	1	1	1	0	1	1
1	0	0	1	1	1	1	1	1

McCluskey:

L_0	2	1	0	L_1	2	1	0
	0	1	0		0	1	-
	1	0	0		1	0	-
	1	0	1		1	0	-
	1	1	0		-	1	0
	1	1	1				

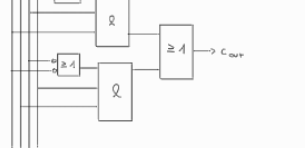


McCluskey:

L ₀	3	2	1	0	L ₁	3	2	1	0	L ₂
	0	1	0	1		1	-	1	0	✓
	1	0	1	0		1	0	1	-	✓
	1	1	1	0		-	1	0	1	✓
	1	1	0	1		0	1	-	1	✓
	1	0	1	1						
	0	1	1	1						

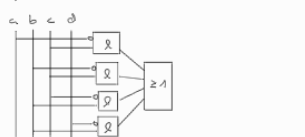
$\Rightarrow x_3 x_2 x_1 x_0 + x_3 x_2' x_1 + x_3 x_2' x_1' x_0 + x_3' x_2 x_1 x_0$

$x_3 x_1 (x_2 + x_2') + x_3 x_0 (x_2 + x_2')$



- 4a)
i) $27_{10} = 0001\ 1011\ 0_2$
ii) $0,3_{10} = 0,01001_2 \rightarrow 0100\ 1100\ 1101\ 10^{-2}$
b) $27_{10} \text{ IEEE: } 0\ 10011\ 1011\ 0000\ 00_2 = 0x4e3d$
 $0,3_{10} \text{ IEEE: } 0\ 01101\ 0011\ 0011\ 00_2 = 0x348d$

5)
 $a+b$: OR
 $a \cdot b$: AND
 $((ab)'c)' \cdot ((cd)'b) = ((ab)'c)' \cdot ((cd)'b)$



x_3	x_2	x_1	x_0	y_3	y_2	y_1	y_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	1	0
0	0	1	0	0	0	1	1
0	0	1	1	0	1	0	0
0	1	0	0	0	0	1	0
0	1	0	1	0	1	0	1
0	1	1	0	0	1	1	0
0	1	1	1	0	1	1	1
1	0	0	0	0	0	0	1
1	0	0	1	0	0	1	0
1	0	1	0	0	0	1	1
1	0	1	1	0	0	1	1
1	1	0	0	0	1	0	0
1	1	0	1	0	1	0	1
1	1	1	0	0	1	1	0
1	1	1	1	0	1	1	1

Umbilder:
 $y_3'(x) = \{1111\}$
 $y_2'(x) = \{0011, 0101, 0110, 0111, 1001, 1010, 1011, 1100, 1101, 1110, 1111\}$
 $y_0'(x) = \{0001, 0010, 0100, 1000, 0111, 1100, 1011, 1101\}$
Ja, ist sie. Aufgrund der Verteilung auf dem KV-Diagramm, welchen die Unidealgte, unigide Verteilung ist, ist es nicht möglich, sie zu vereinfachen.

- 1)
 $A = \{0, p\}, 0 < p$
a) $|A^c| = 32$
b) $H = \{000pp, 00p0p, 00pp0, 0p00p, 0p0p0, 0pp00, p000p, p00p0, p0p00, p0000\}$
c)
 $1 \bmod 11 = 1$
 $2 \bmod 11 = 2$
 $3 \bmod 11 = 3$
 $4 \bmod 11 = 4$
 $5 \bmod 11 = 5$
 $6 \bmod 11 = 6$
 $7 \bmod 11 = 7$
 $8 \bmod 11 = 8$
 $9 \bmod 11 = 9$
 $10 \bmod 11 = 10$
d) $\{(1, 11), \dots, (5, 11)\}$

- 3)
a) i) $a: [0, 63], b: [0, 2^{10}], c: [0, 15]$
ii) $a: [-32, 31], b: [-2^{10}, 2^{10}], c: [-8, 7]$
b) $w = 1010\ 0100\ 1111\ 1100\ 0000\ 0001\ 0111\ 0001$
i) $a = 5, b: 1025, c = 8$
ii) $a = 5, b: -1023, c = -8$
c)
 $w = 00011111110000111111111100011110$
d)
a) i) $0,1875_{10} = 0,0011_2 = 1 \cdot 10^{-3}$
 $0,375 = 0$
 $0,75 = 0$
 $1,5 = 1$
 $1,0 = 1$
ii) $24 \frac{1}{2} = 0001\ 1000, \overline{001}_2 = 1,1 \cdot 10^{-2}$
 $\frac{2}{7} = 0$
 $\frac{1}{4} = 0$
 $\frac{1}{2} = 1$
 $\frac{2}{7} = 0$

- 1) a) i) $5^2 = 125$ ii) $5^3 = 675$
b) $nqz\bar{s}, n\bar{x}\bar{e}p, p\bar{p}q\bar{r}, p\bar{q}p, \bar{s}p\bar{n}q$
c) $x\bar{z}\bar{x}p, x\bar{z}xq, \bar{x}\bar{z}\bar{x}\bar{x}, \bar{x}\bar{z}\bar{x}\bar{x}\bar{z}\bar{n}, x\bar{z}\bar{x}p, x\bar{z}\bar{x}q, x\bar{z}\bar{x}\bar{x}, \bar{x}\bar{z}\bar{x}\bar{x}, \bar{x}\bar{z}\bar{x}\bar{x}\bar{p}$
d) $\bar{x}_3 \bar{x}_2 \bar{x}_1 \bar{x}_0 = 4003_{10} \rightarrow 2nnx$
e) $n\bar{x}p\bar{q} = 03125, p\bar{s}\bar{s}n = 14405$
 $\frac{1440}{312} = 4,61538$
 $\frac{1133}{1133} + 1$ (Invertieren)
 $\Rightarrow 1134_{10}$ Wertur $\Rightarrow 125_{10} \cdot 25, 3 \cdot 125, 4 \cdot 125 = 198$ Wertur

- 2)
a) 1035 und 224
b) $255 \rightarrow 8, 1024 \rightarrow 11$
 $2000 \rightarrow 11, 768 \rightarrow 10$
c) 128 und -128 :
 $128_{10} \rightarrow 0\ 1000\ 0000_2$
 $-128_{10} \rightarrow 1\ 1000\ 0000_2$
d) $128_{10} \rightarrow 0000\ 1000\ 0000$
 $-128_{10} \rightarrow 1000\ 1000\ 0000$
3) $0x3f = 1001\ 1111_2$
 $0xb0 = 1011\ 0000_2$
 $0x00 = 0000\ 0000_2$
i) $0x3f - 1 = 1001\ 1110$
 $\frac{2\ 1001\ 1111}{1001\ 1110} = 0x5e$
ii) $0xb0 - 1 = 1010\ 1111$
 $\frac{2\ 1011\ 0000}{1010\ 1111} = 0xa0$
iii) $0xb0 \cdot 1 = 1111\ 1111$
 $\frac{9\ 0000\ 0000}{00000000} = 0x00$

b) wenn viele 3a) ii): Die Bedingung $(a \cdot 1)$ zandert jeden Wert $x = 0$ von rechts nach links zu einer 1, bis man auf eine 1 stot, die wird zur 0. Das funktioniert durch den Carry. Durch die 2 Operation werden bits verglichen, sobald alle geanderten bits zu einer 0 werden. Danach folgt, dass von rechts aus jeweils alle 0er bleiben, und sich nur die erste 1 zu einer 0 andert.

c) $y(x_3 x_2 x_1 x_0)$
KV:

	00	01	11	10
00		1		1
01	1		1	
11		1		1
10	1		1	

 $\{0011, 0101, 0110, 0111, 1001, 1010, 1011, 1100, 1101, 1110, 1111\}$
 $x_3 x_1 = x_3 x_0 + x_2 x_0 + x_2 x_1 + x_3 x_2 + x_1 x_0$
Also Primimplikanten, alle Erennen
i) $x_3 x_0 + x_2 x_0 + x_2 x_1 + x_3 x_2 + x_1 x_0$
Da das auchsch komplett aus existierenden Primimplikanten besteht, ist es der einzige mogliche, der korrekt ist.

1)
a) $3'' = 81_{10}$
b) $W_{kinetik} = 0$
 $y_{\text{rohr}} = PPPP = 80_{10}$
c) $NAN, NMP, NNM, \underline{NNN}, NNP, NPM, NPN$
2)
a)

$\frac{138}{730}$	$\frac{538}{842}$	$\frac{E0A}{FAC}$	$\frac{ECE}{753}$	$\frac{5FC}{A0F}$
875	DDA	E86	621	406
carry		carry	carry	carry
		overflow	overflow	overflow

b) $FAC < EDA < ECE < ADF < 8FC$
 $< 8A2 < 138 < 538 < 730 < 753$

3a) i) $a: [0, 127], b: [0, 511], c: [0, 31]$
ii) $a: [-64, 63], b: [-256, 255], c: [-16, 15]$
b) $w = 1110\ 0000\ 1010\ 1000\ 0110\ 0111\ 1000\ 0000$
a b c
i) $a = 65, b = 25, c = 16$
ii) $a = -63, b = 25, c = -16$
c) $c: 0011001$
 $b: 11111111$
 $c: 111100$
 $\Rightarrow w = 00001100100010111111100011100000$
 $w = 0C87FCE0$

4)
a) i) $30,0_{10} = 0011\ 0010,0_2 = 1,100 \cdot 10^2$
ii) $0,05_{10} = 0,000011_2 = 1,1 \cdot 10^{-5}$
b) $10\ 000\ 0000\ 0000\ 0000\ 0011_2$
benotigt Stellen: 25_{10}
 $10\ 0000\ 0000\ 0000\ 0000\ 0010 \cdot 2^{25}$
 $= 0100\ 0011\ 0000\ 0000\ 0000\ 0000\ 0010_2$
 $= 0x43000002$
5) $y^{-1}(1) = \{0x0, 0x7, 0xE\}$

b) $x_3 x_2 x_1 x_0 + x_2 x_1 x_0 + x_1 x_0 x_0$
c)

6) $\{((a+b) \cdot (c+e))' \cdot ((c+e)' \cdot d \cdot f)'\}$
 $a'b + ae' + c \cdot e' + d' + f$
 $\rightarrow c'b + c + d' \cdot e' + f$
g: $a' + b + c + d' + e' + f$
b) Nein, sind Sie nicht.
fur: 110110 ist g richtig, aber f nicht.
7) $g^{-1}(1) = \{4, C, F, G, D, 9, A, B\}$
a)

	00	01	11	10
00				
01	1	1	1	1
11				
10		1	1	

Primimplikanten: $\{x_2 x_1', x_2' x_1, x_3 x_2', x_3' x_2, x_3 x_1', x_3' x_1, x_3 x_0', x_3' x_0\}$
alle existieren, außer $x_2 x_0$

c) alle auer $x_2 x_1'$ und $x_1' x_2$
d) $x_2 x_1' + x_1' x_2 + x_3 x_2' + x_3' x_2 x_0$
e) Ja: $x_3 + x_3 x_2' x_0$
7) KV:

	00	01	11	10
00				
01	1	1	1	1
11				
10		1	1	

essentielle Primimplikanten: $\{x_2 x_1', x_1' x_2, x_3 x_2', x_3' x_2\}$
nicht es: Primimplikanten: $\{x_3 x_1', x_3' x_1, x_3 x_0', x_3' x_0\}$
b) $x_1' x_2' + x_1' x_2 + x_2 x_1' + x_2 x_1$
und
 $x_1' x_2' + x_1' x_2 + x_2 x_1' + x_2 x_1$
c) Ja: $x_3 x_1' + x_3 x_1$

- 1) $|A^4| = 3^4 = 81$
- b) $\text{Wenk: } \text{MMMM} \equiv 0$
 $\text{Jupk: } \text{PPPP} \equiv 80$
- c) $\text{NMN, NMP, VNM, VNN, NNP, NPM, NPN}$
 (aufsteigend)

2)

a)

138	538	E0A	ECE	5FC
+ 73D	+ 8A2	+ F AC	+ 753	+ A0F
875	D04	E86	62A	40B
carry	carry	carry	carry	carry
		overflow	overflow	overflow

- b) $8A2 < 5FC < A0F < ECE < EDA < 7AC < 138 < 538 < 73D < 753$

- 3)
- a) i) $a: [0; 127], b: [0; 311], c: [0; 31]$
- ii) $a: [-64; 63], b: [-256; 255], c: [-16; 15]$

b) $w = \underbrace{1110 \ 0000 \ 1010 \ 1000 \ 0110 \ 0111 \ 1000 \ 0000}_c$

- i) $a = 65, b = 25, c = 16$
- ii) $a = -63, b = 25, c = -16$

c)

$w = 0000 \ 1100 \ 1000 \ 0111 \ 1111 \ 1100 \ 1110 \ 0111$
 $w = 0x0C87FCE7$

4)

i) $50, 0_{\infty} = 0011 \ 0010, 0_2$
 $\Rightarrow 1,1 \cdot 10^6$

ii) $0,05_{\infty} = 0,000011_2$

0,1	0
0,2	0
0,4	0
0,8	0
1,6	1
1,2	1

$\Rightarrow 1,1 \cdot 10^{-6}$

4b)

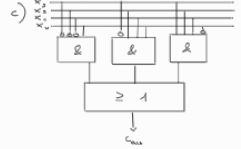
unsignael: 1 0000 0000 0000 0000 0000 001A₂
 benötigte Stellen: 25

unsignelisiert: 1 · 10²⁴

IEEE binary: 0100 1011 1000 0000 0000 0000 0000
 IEEE hex: 0x4B800000

5) $\gamma^{-1}(1) = \{0, 7, E\}_{46}$

b) DNF: $x_3'x_2'x_1'x_0' + x_3'x_2x_1x_0 + x_3x_2x_1x_0'$



6)

a) $f: ((a \cdot b)' \cdot (c' \cdot e))' + ((e' \cdot e)' \cdot d \cdot f)'$
 $= a'b + ae' + c \cdot e' \cdot d' + f \neq ab + c \cdot d' \cdot e' \cdot f$
 $g: ((a+b)' \cdot c')' + ((e' \cdot f)' \cdot d)'$
 $= a + b + c + d + e' + f \neq a'b + c + d' \cdot e' + f$

b) $\Rightarrow$ Nein, die Abbildungen f & g sind nicht invertierbar.
 Sie unterscheiden sich z.B. für:
 $100110: f(100110) = 1$
 $g(100110) = 0$

7)

a) $g^{-1}(1) = \{4, C, 6, 7, A, B, D\}$

b)

	00	01	11	10
00				
01	1	1	1	1
11	1	1	-	-
10	-	-	1	1

essentielle P: $\{x_3'x_2, x_3x_2', x_3x_1'x_0\}$
 nicht ess. P: $\{x_3x_2'x_0, x_3x_1'x_0\}$

c) 5, c

alle außer x_2x_1' und $x_3'x_0$

d)

$x_2'x_0 + x_3x_2' + x_3x_2x_0 + x_3x_2'x_0$

e)

$\text{Ja, gibt es: } x_2 + x_3x_2 + x_3x_0$

1)

a) $A05E < A0E5 < A50E < AE30 < E05A < 0A5E < 50AE$

b)

A05E	A05E	A50E	E05A
- 0A5E	- 50AE	- A05E	- A05E
5600	4FBD	04BD	4FFB
	overflow	overflow	overflow

A05E	A05E
- A05E	- A05E
00F2	0087
overflow	overflow

c) v)

$00F2 \bmod 7 \equiv 0$
 $d.h. D + F + 2 \bmod 3 \equiv 0$
 $D + F + 2 \bmod 5 \equiv 0$

$00F2 \bmod E \equiv 0$
 $d.h. 2 \bmod 2 \equiv 0$
 $2 \cdot DF - 4 \bmod 5 \equiv 0$

vi)

$8 \cdot 7 \bmod 7 \equiv 0$
 $87 \bmod 3 \equiv 0$

2)

a)

i) $x = 1010 \ 1000 \ 1011 \ 0101$
 $t = x \ll 6 = 0010 \ 1101 \ 0100 \ 0000_2 = 2B40_{16}$
 $y = t \gg 11 = 0000 \ 0000 \ 0000 \ 0101_2 = 0005_{16}$
 $z = y \gg 11 = 0000 \ 0000 \ 0000 \ 0101_2 = 0005_{16}$

ii) $x = 0011 \ 0111 \ 1111 \ 0000$
 $t = x \ll 6 = 1111 \ 1100 \ 0000 \ 0000_2 = 0xFC00$
 $y = t \gg 11 = 0000 \ 0000 \ 0001 \ 1111_2 = 0x001F$
 $z = t \gg 11 = 1111 \ 1111 \ 1111 \ 1111_2 = 0xFFFF$

b)

i)

$t = +256 \cdot 16 \cdot 2 + 256 \cdot 11 + 64_{10}$
 $y = +5_{10}$
 $z = +5_{10}$

ii)

$t = -4 \cdot 16^3 - 16^2 - 16^1$
 $y = +31$
 $z = -1$